

GRAIN QUALITY ASSESSMENT CHECKLIST

Static Test Evaluation Protocol for KNSB Propellant Grains

Date of Preparation		Time of Preparation	
Prepared By (Technician)		Grain / Batch ID	

1. MATERIAL FORMULATION & COMPOSITIONS

PARAMETER DESCRIPTION	TARGET / SPECIFICATION	ACTUAL MEASUREMENT	VERIFIED
Ratios of KNSB (KN : Sorbitol)	65 : 35 (Standard)		☐
Weight of Potassium Nitrate (KN)	g	g	☐
Starting Weight of Sorbitol	g	g	☐
Weight of Sorbitol (After Saturation/ Dehydration)	g	g	☐
Total Weight of KNSB Propellant	g	g	☐

2. CASTING & PROCESSING PARAMETERS

PARAMETER DESCRIPTION	TARGET / SPECIFICATION	ACTUAL MEASUREMENT	VERIFIED
Pressure Used When Casting	psi / MPa	psi / MPa	☐
Density of KNSB Grain	Nominal: ~1.84 g/cm³	g/cm³	☐

3. LINER & GRAIN GEOMETRY DIMENSIONS

PARAMETER DESCRIPTION	TARGET / SPECIFICATION	ACTUAL MEASUREMENT	VERIFIED
Weight of Liner	g	g	☐
Dimensions of Liner: Length	mm	mm	☐
Thickness of Liner	mm	mm	☐
Outer Radius of Grain	mm	mm	☐
Inner Radius of Grain (Core Radius)	mm	mm	☐

4. SUPERVISOR REVIEW & SIGN-OFF

Supervisor
Name: _____
Print Name

Signature: _____
Sign Manual/Electronic

Approval
Date: _____
DD / MM / YYYY

Status: ☐ Approved for Static Test ☐
Rejected

Certification Statement: By signing above, the supervisor verifies that the structural integrity, dimensional alignment, mass properties, and casting variables of the stated KNSB propellant grain batch have been inspected and comply with the required static testing standard operating procedures.